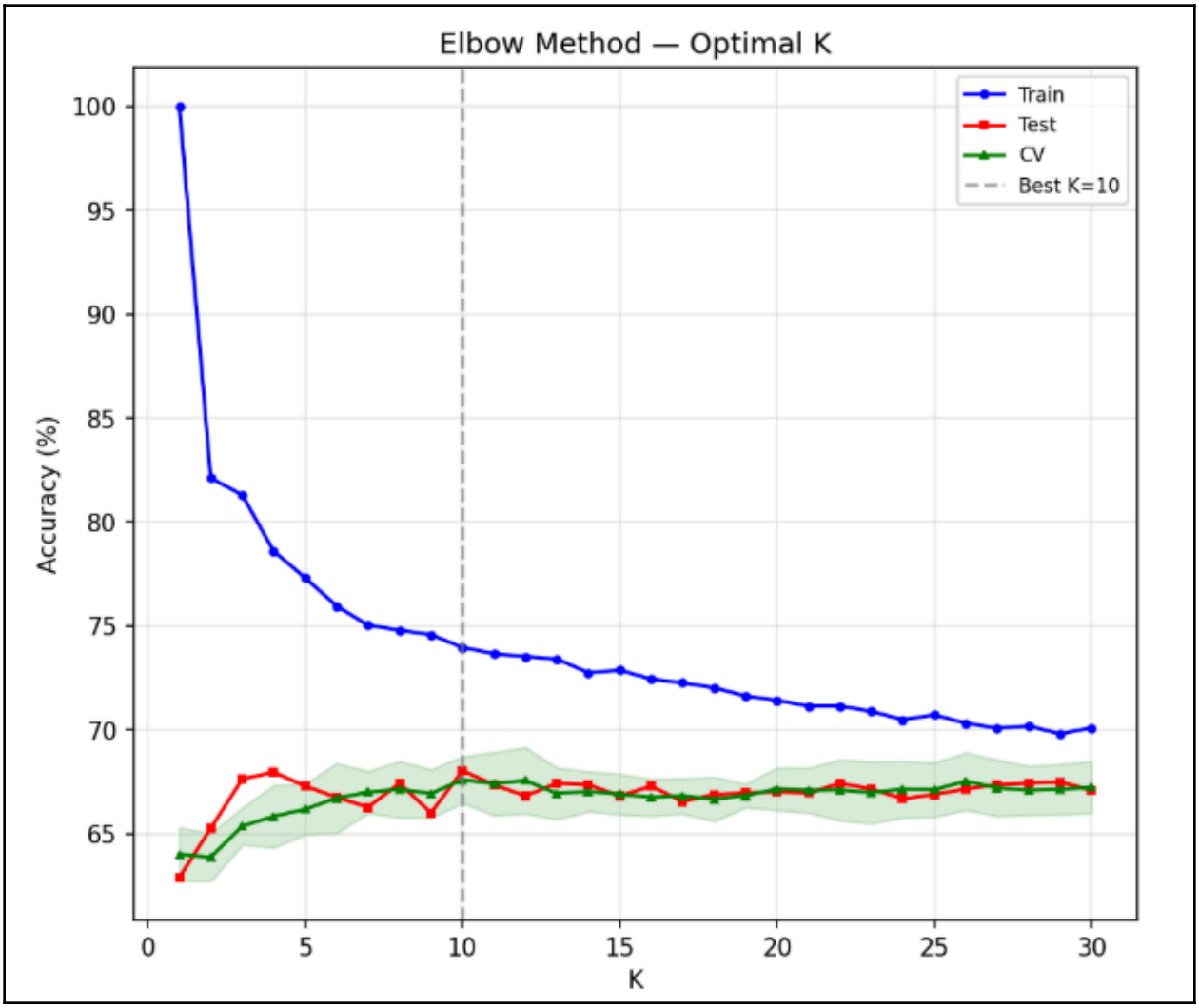
Experiment 4

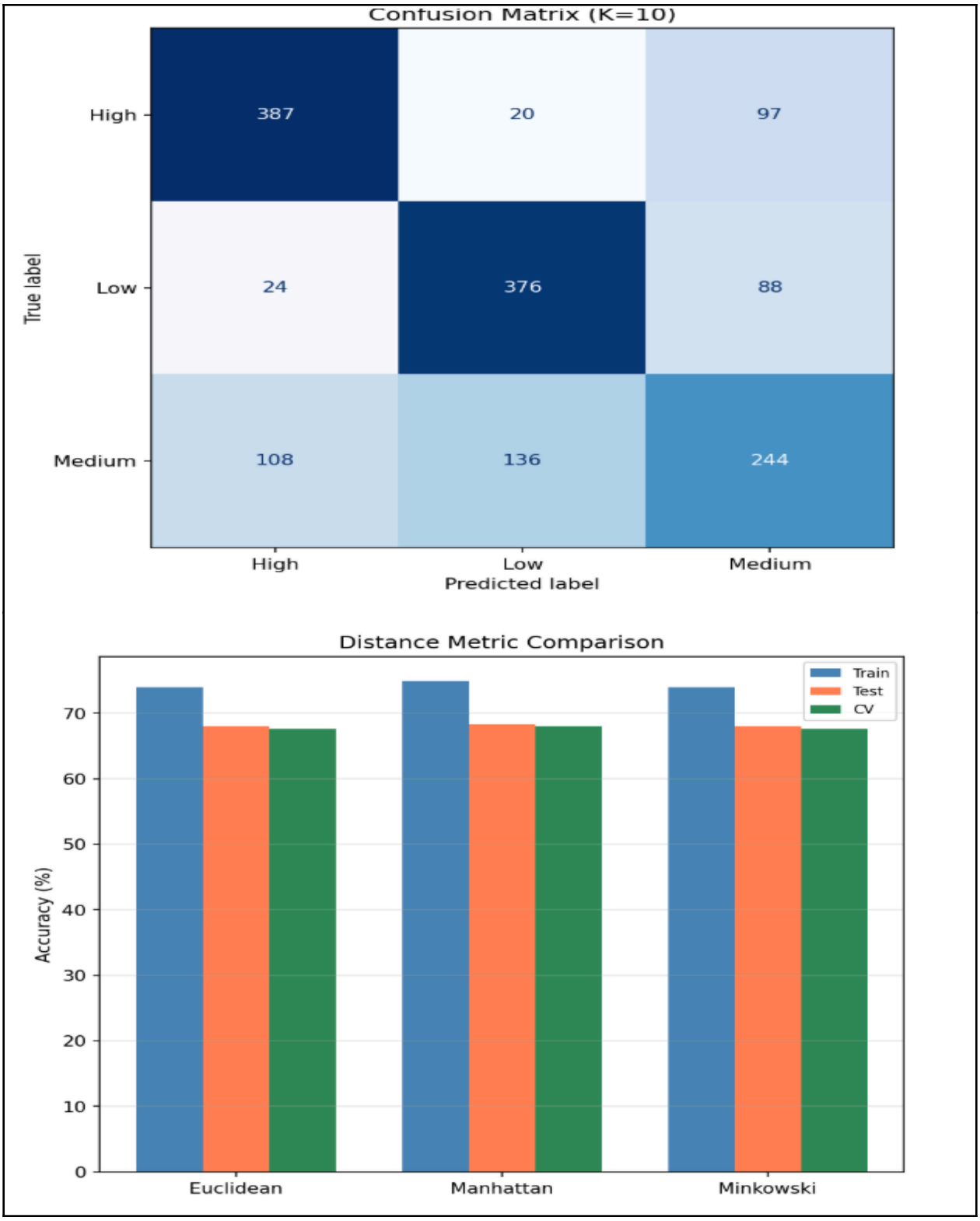
Aim : Implement K-Nearest Neighbors (KNN) and evaluate model performance

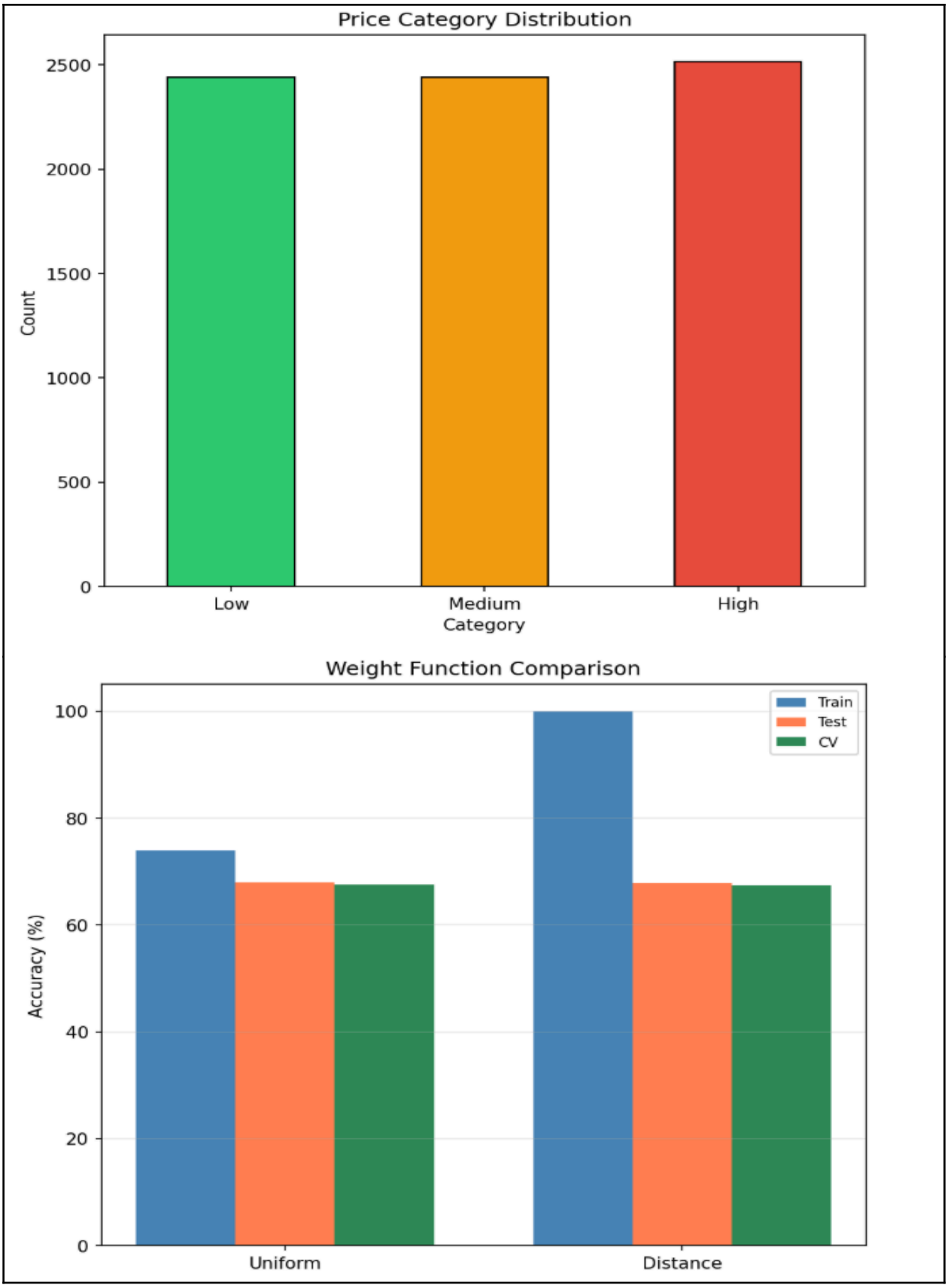
Data Source : (Used Car Price Prediction Dataset)

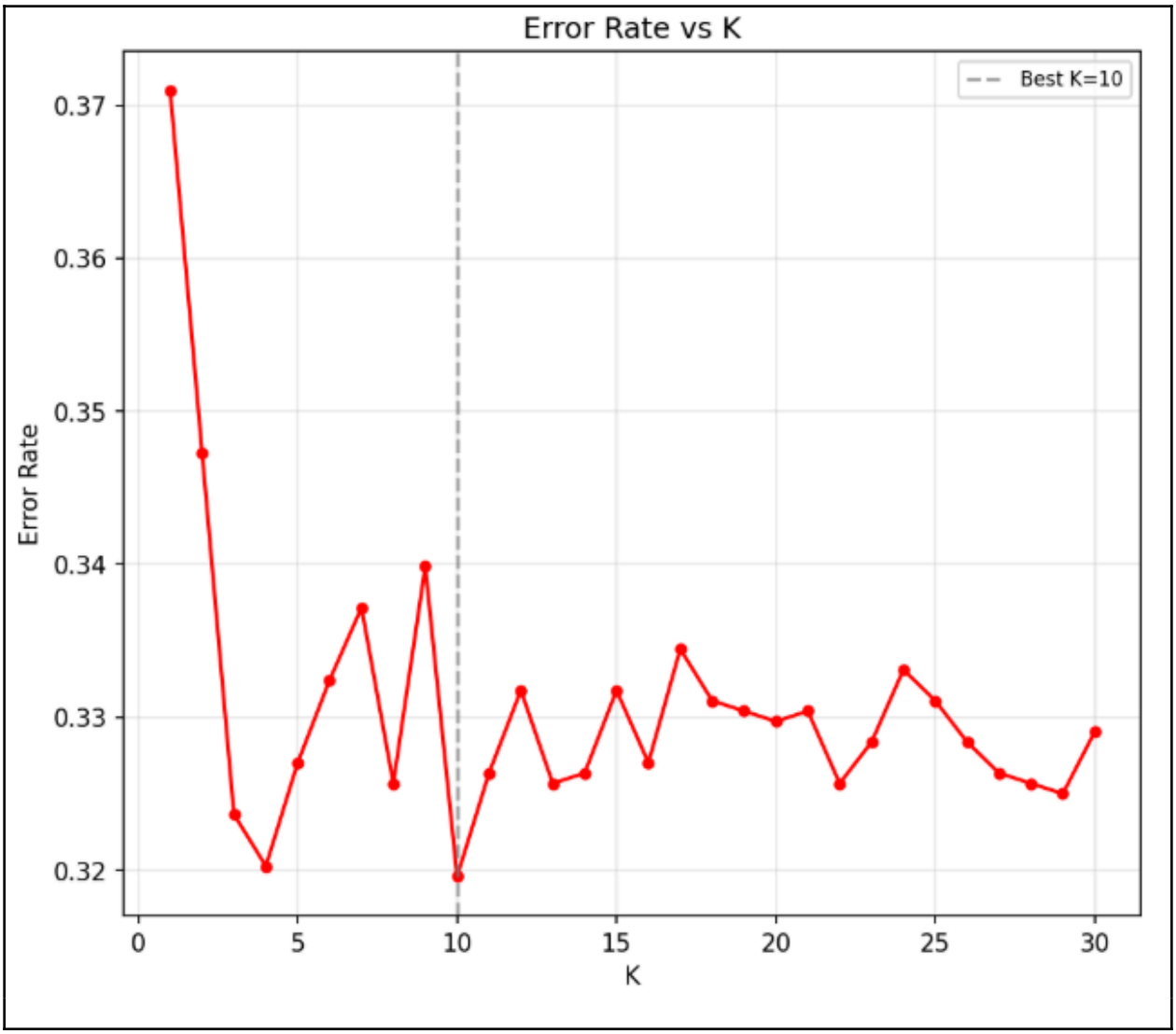
- Dataset Type: Structured tabular dataset
- File Format: CSV (Comma Separated Values)
- Total Records (Rows): 7,400
- Total Attributes (Columns): 29
- Domain: Used Car Sales / Automobile Marketplace Data
- Dataset Link

Output :




Confusion Matrix (K=10)
High
Low
Medium
True label
387
20
97
24
376
88
108
136
244
High
Low
Medium
Predicted label
Distance Metric Comparison
Train
Test
CV
Accuracy (%)
0
10
20
30
40
50
60
70
Euclidean
Manhattan
Minkowski


Price Category Distribution
Count
2500
2000
1500
1000
500
0
Low
Medium
High
Category
Weight Function Comparison
Accuracy (%)
100
80
60
40
20
0
Train
Test
CV
Uniform
Distance



Conclusion :
K-Nearest Neighbors (KNN) is a simple yet powerful classification algorithm that makes predictions based on similarity measures. It does not require model training but depends heavily on the choice of K and proper feature scaling. Model performance is evaluated using metrics such as accuracy, precision, recall, F1-score, and confusion matrix. Although computationally expensive for large datasets, KNN performs effectively for smaller and well-scaled datasets.